Homework 1: Image Enhancement

Computer Vision – Fall 2026

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1 Objective and data

The goal is to build and compare a pipeline of classical methods for enhancing color images affected by low contrast, non-uniform illumination and noise, and to evaluate the trade-offs between detail visibility, noise and natural appearance.

Images. Six color photographs were used, each showing at least one of the required conditions: `badcats` (dark indoor scene, two cats on a table), `baddesktop` (underexposed desk with various papers and objects), `guyinthefog` (backlit subject against a foggy sky), `girlinthesnow` (washed-out, low-contrast close-up in the snow), `goat` and `schoolkids` (scans of faded old prints with a color cast). **Source and license:** personal photographs taken by me, redistributed in the repository with permission of the people depicted. All originals and degraded versions are included in the repository. For two images (badcats and baddesktop) I also provided a "perfect" version (perfectcats and perfectdesktop).

Synthetic degradation. To make the comparison reproducible, each original I is degraded by (1) reducing its contrast towards the global mean μ of the image, $I' = \alpha(I - \mu) + \mu$ with $\alpha = 0.5$, and (2) adding zero-mean Gaussian noise with $\sigma = 8$ (on the 0–255 scale), generated with `numpy.random.default_rng` and a fixed seed of 42; the result is clipped to $[0, 255]$. Since the same α is applied to all three channels, the degradation also halves the differences between channels, i.e. it reduces color saturation.
The original images serve as references for evaluating the synthetic degradations only; in real-world scenarios a clean reference is usually not available. Note also that several originals are themselves dark or low-contrast, so they are a reference, not an ideal target.

2 Methods and parameters

All methods are applied to the same degraded images and evaluated with the same metrics.

1. **Baseline**: the degraded image without processing.
2. **Linear contrast stretching**: intensities between the 1st and 99th percentile ($r_{\min}, r_{\max}$) are mapped linearly to $[0, 255]$, $s = 255\,(r - r_{\min})/(r_{\max} - r_{\min})$, clipping the rest (percentiles make it robust to noisy outliers). Two variants: **(a) per channel**, on B, G, R independently, and **(b) luminance only**, on the Y channel of YCrCb with Cr, Cb unchanged. Variant (a) uses different gains per channel, so it changes the color balance; variant (b) preserves hue.
3. **Histogram equalization** (`cv2.equalizeHist`) applied only to a luminance-like channel to avoid hue changes: **(a)** the L channel of CIE L*a*b* and **(b)** the V channel of HSV ($V = \max(R, G, B)$, H and S unchanged).
4. **Own method – denoising + stretching + chroma gain** (motivated combination): (i) non-local means denoising (`cv2.fastNlMeansDenoisingColored`, $h = 3$, $h_{\text{color}} = 9$, template window 7, search window 21) *before* any contrast increase, so that the noise is not amplified; (ii) linear stretching of the L channel of L*a*b* (percentiles 1–99); (iii) chroma gain: $a \leftarrow 128 + g(a - 128)$, $b \leftarrow 128 + g(b - 128)$ with $g = 2$. Scaling a and b by the same factor keeps the hue angle and restores the saturation lost in the degradation ($g = 1/\alpha$). The drawback is that denoising also smooths fine details and texture.

Metrics, computed for each degraded/processed image with respect to the original: **PSNR** $= 10 \log_{10}(255^2/\text{MSE})$ (own implementation) and **SSIM** (`skimage.metrics.structural_similarity`, computed per channel and averaged, data range 255). Higher is better for both.

3 Results

Figure 1 shows the visual comparison and Table 1 the metrics. Figure 2 shows a zoomed detail.

General observations. *Linear stretching per channel* gives the strongest contrast and the most vivid colors, but it shifts the white balance because each channel gets a different gain: the sky in `guyinthefog` becomes cyan/pink and the sky in `goat` becomes an unnaturally saturated blue. *Stretching on Y* keeps the hues but the images look grayish, because the chroma reduced by the degradation is left untouched. *Histogram equalization*

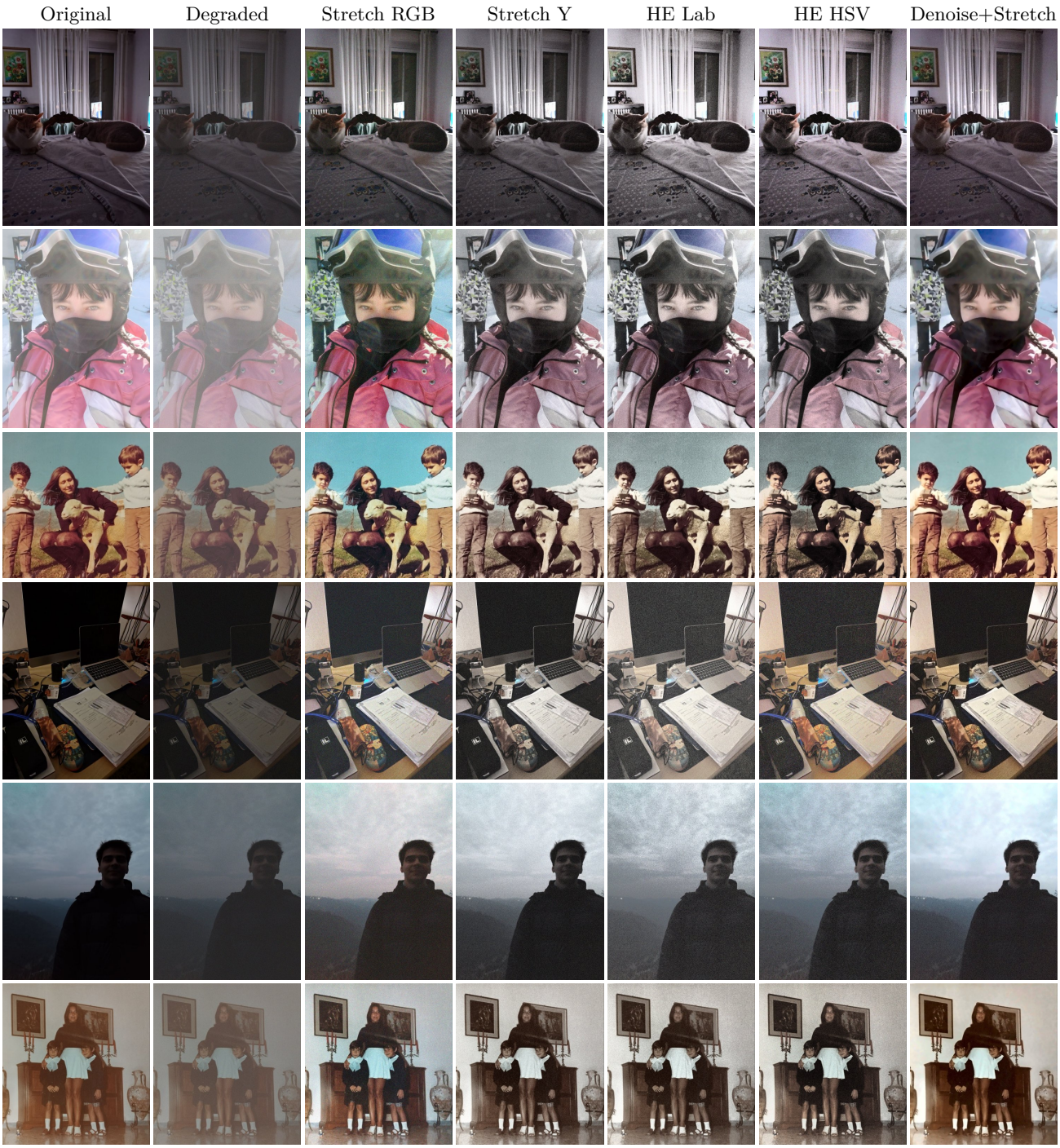
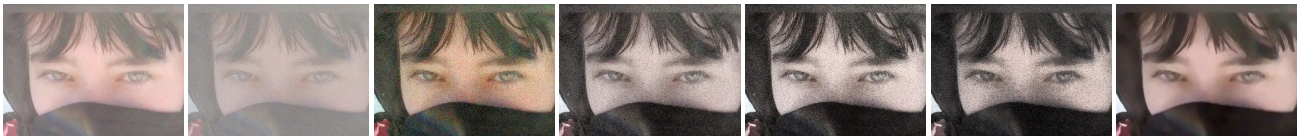


Figure 1: Visual comparison for all images and methods. Each row: original (reference), degraded (= baseline) and the results of the five processed variants, all shown with the same display scale (8-bit, no further normalization). Full-resolution side-by-side figures are in the repository (`results/`).

Image	Baseline	Stretch RGB	Stretch Y	HE Lab	HE HSV	Denoise+Stretch
`badcats`	**21.23** / **0.520**	18.72 / 0.285	20.29 / 0.400	10.59 / 0.206	11.75 / 0.196	19.60 / 0.415
`girlinthesnow`	**20.96** / 0.538	13.74 / 0.207	13.95 / 0.289	12.80 / 0.237	12.21 / 0.222	13.04 / **0.715**
`goat`	**19.58** / 0.517	17.16 / 0.250	17.33 / 0.357	16.52 / 0.300	17.26 / 0.275	18.28 / **0.736**
`baddesktop`	**20.68** / **0.363**	11.50 / 0.132	12.09 / 0.179	8.87 / 0.152	10.53 / 0.119	12.45 / 0.191
`guyinthefog`	**18.10** / 0.355	12.61 / 0.098	12.38 / 0.161	12.97 / 0.092	14.41 / 0.095	11.07 / **0.398**
`schoolkids`	**20.33** / 0.506	16.75 / 0.178	16.85 / 0.260	16.78 / 0.239	16.90 / 0.199	17.05 / **0.649**
Mean	**20.15** / 0.467	15.08 / 0.192	15.48 / 0.274	13.09 / 0.204	13.84 / 0.184	15.25 / **0.518**

Table 1: PSNR [dB] / SSIM of each method with respect to the original image (higher is better). Best value of each row in bold. The same values are in `results/metrics.csv`.



Original Degraded Stretch RGB Stretch Y HE Lab HE HSV Denoise+Stretch

Figure 2: Zoomed detail of `girlinthesnow` (face): equalization strongly amplifies the noise, the own method removes it but smooths the skin texture.

gives the largest increase of local contrast and reveals the most details in dark regions (curtains and fur in `badcats`, the keyboard and papers in `baddesktop`), but it is also the method that amplifies the noise the most (Fig. 2): the gray levels of large uniform regions (sky, walls, snow) are spread over a wide range, so the grain becomes very visible. The Lab and HSV versions are similar; HSV tends to keep more saturated colors, while Lab gives slightly more natural tones. Both have the lowest PSNR (mean 13.1 and 13.8 dB). *Denoise+Stretch* gives the cleanest images and the most natural colors, at the cost of smoothed textures (skin, fur, grass).

When better numbers do not mean a better image. The baseline has the *highest PSNR for every image* (mean 20.2 dB), yet it is clearly the worst image visually: flat, gray and noisy. The degradation preserves the mean of the image, so the pixel-wise error stays moderate, while every enhancement changes the brightness and contrast with respect to the original (and amplifies the noise), increasing the MSE. PSNR therefore rewards "doing nothing". In `baddesktop` the problem is even stronger: the original itself is underexposed, so all the methods that make the papers, the keyboard and the objects on the desk more visible move *away* from the reference and lose 8–12 dB of PSNR. The two metrics also disagree with each other: on `girlinthesnow` the own method has the best SSIM (0.715, against 0.538 for the baseline) but a PSNR 7.9 dB lower than the baseline. SSIM is more sensitive to the noise (which destroys local structure) and less to a global change of brightness, so it matches visual judgment better here, but it is not perfect either (see `guyinthefog` below). For these reasons no method was chosen on a single metric.

4 Qualitative analysis of selected cases

Case 1 – `girlinthesnow`. *Preferred method: Denoise+Stretch.* It recovers the contrast of the face, the dark mask and the helmet and brings back the original colors of the pink jacket and the blue goggles, without visible noise; it also has the highest SSIM (0.715). The details that became easier to see are the eyes, the hair under the helmet and the patterns on the jacket of the person in the background. Trade-offs: the skin and the snow are smoothed (loss of fine texture, Fig. 2) and the jacket is slightly more saturated than in the original. Stretch RGB is a good second choice for contrast, but it adds a bluish cast to the snow. Both equalizations make the face grainy and turn the jacket into a dull gray-pink.

Case 2 – `guyinthefog`. Equalization (especially HSV, the best PSNR among the processed results, 14.4 dB) makes the face and the clothes of the subject visible, but the uniform foggy sky is transformed into strong grain. The own method has the best SSIM (0.398) and no noise, but the global stretch on L pushes the sky to almost white (clipping) and the cyan tint of the sky in the highest part is exaggerated by the chroma gain; it also has the lowest PSNR (11.1 dB). Stretch RGB creates a color cast in the sky. *Preferred method: Stretch Y*, as a compromise: the face is more readable than in the degraded image, colors stay natural and the noise is amplified less than with equalization. None of the global methods handles this scene well.

Case 3 – `badcats`. In this dark scene Stretch Y is a good compromise (second best PSNR, 20.3 dB): the cats, the painting and the folds of the tablecloth become visible without color shifts. Equalization reveals more details in the curtains and the fur, but with a strong grain and an unnatural, too bright look. Here the own method also

gives good colors and makes the details in the tablecloth more visible, but the denoising flattens the fur of the cats.

Case 4 – `schoolkids`. In This image has both the best PSNR among the processed results (17.05 dB) and the best SSIM (0.649, against 0.506 for the baseline). *Preferred method: Denoise+Stretch.* The wall and the floor are clean, the white details in the clothes have become more visible, and the faces, the candlesticks, the framed pictures and the vase easier to see. Its main trade-off is the color: the chroma gain brings back the warm tones of the print but exaggerates them a little, so the skin and the wood become a bit too orange; the legs and the uniforms also lose some texture because of the denoising. *Another good method: Stretch RGB.* It gives the most vivid and contrasted image, but the independent gains remove the warm cast of the print and turn the whites (the skirt, the bows, the wall) bluish, and the grain is clearly visible on the wall. Stretch Y keeps the hues but leaves the image almost sepia and noisy. Both equalizations make the photo look nearly black and white, with strong grain on the uniform wall.

5 Conclusions and limitations

No method is best on all images and criteria. Histogram equalization maximizes the visibility of details but amplifies the noise; per-channel stretching gives vivid colors but changes the white balance; luminance-only methods preserve hues but leave the colors dull. Denoising before stretching, plus a chroma gain that compensates the known degradation, gives the most natural results and the best mean SSIM (0.518), at the cost of smoothed textures and possible clipping in very bright regions. Limitations: all methods are global, so they cannot correct strongly non-uniform illumination; the parameters were fixed for all images and not tuned per image; the metrics need a clean reference, which is not available in real scenarios.

Repository: `https://github.com/Ludokmain/Image_enhancement` – Python source code, original images and all the result will be available after the command (`python project.py`). Refer to the README.md for instructions.